

A FRAMEWORK FOR SOCIAL MOTIVES¹

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In any complex social system, the choices of a single person often affect the outcomes of others. When one takes the others' outcomes into account in making choices, we say one is manifesting a social motive. We assume that an individual's social motives are manifested in his social preferences. Any theory of decision or choice requires information about preferences, and so in addition to attempting to clarify definitions of social motives, this study provides a framework for theories of social decision.

To begin, we establish operational definitions for the motives of self-interest, self-sacrifice, altruism, aggression, cooperation, and competition. These definitions are based on the simple operators of summations and differences. Then we examine some supplementary motives involving proportionality operators. Next we eliminate the assumption that an individual has a fixed preference structure which is applied to all social choices. This leads to a focus on the specific distribution of consequences resulting in conditional motives; conditional motives reflect varying basic motives depending on whether the individual is ahead or behind. All these motives are represented graphically, and a correspondence matrix is given to illustrate the interrelationships among the motives.

Finally, we investigate combinations of the motives since such composites are less restrictive and can better account for observed behavior. Linear combinations are interesting but still too restrictive. Conjunctive, disjunctive and lexicographic combinations offer useful possibilities for characterizing particular social motives. Complex combinations involving general nonlinear forms arise and some representative forms are explored.

INTRODUCTION

IN ANY COMPLEX social system characterized by a high degree of social interdependence, one's acts often have direct consequences for the outcomes of other members of the system. When an individual takes the outcomes of others into account in making a choice, we say that one is manifesting a social motive. There are, of course, many subtle and complicated ways in which others' outcomes may be taken into account and, while we will illustrate some of these, our primary goal in this paper will be to develop and explore a framework for social motives which can aid our understanding of social behavior as well as the social system in which we live. More than a decade ago, an important text on motivation stated that "one task for a psychology of motivation might be to provide a useful scheme for classification of

specific wants that have much in common" (Atkinson, 1964, p. 6). In this paper, building on the work of McClintock (1972) and Griesinger and Livingston (1973), we shall attempt to specify precise meanings for some important social motives. Then we shall identify some of the interrelationships among, and implications of, these definitions.

Our fundamental assumption is that motives are manifested in preferences, and that different motives lead to different patterns of preference. Hunger, for example, is manifested by a preference for food or eating over other stimuli or activities; altruism is manifested by a preference for helping over hurting or neglecting. Although some of our discussion is applicable to animal behavior, we explicitly focus on the motives and preferences of humans making conscious choices since it is the

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human social system with which we are mainly concerned.

For any study of decision or choice, we need information about the decision maker's preferences. We need information about a decision maker's social preferences in cases when that choice is based on the consequences it has for others, or when the choices of a number of decision makers mutually determine the consequences for one another. We can ignore social preferences only in cases when a single, self-interested decision maker makes choices in a situation he completely controls.

Empirical studies and casual observation indicate conclusively that individuals are seldom concerned only with consequences to themselves; in one way or another they are also concerned with the progress of others. Even the view of a self-interested, economic man can include social considerations. Economic theory has not ruled out social motives; it has just ignored them. As Arrow (1967, p. 124) notes,

We presume that each individual has some way of ranking social actions according to his preferences for consequences. These preferences constitute his value system. They are assumed to reflect already in full measure altruistic or egoistic motivations as the case may be.

If, rather than assuming that an individual has a fixed preference structure which is applied to all social choice situations, we could discover how the relative distribution of consequences or payoffs determines his preferences, we would have a better basis for understanding social behavior.

In societal decisions, most alternatives have quite different consequences for the social units involved. One type of tax policy, for example, will benefit the poor at the expense of the rich, another type will benefit the married at the expense of the single, etc. Whether we are dealing with a democratic, representational, or dictatorial situation, an understanding of social motives of the decision makers provides us with a basis for predicting or improving social choice. In dictatorial situations,

knowing the dictator's social motives will allow us to predict the choices in situations not yet observed. In democratic or representative situations in which individual orderings are combined into a social decision function, knowing the decision makers' social motives would be useful in establishing rules for aggregating individual preference orderings. For example, even the Pareto principle—the sacred cow of social choice—can be criticized when we examine such social interactions as excessive nosiness (Sen, 1970, sec. 6.5).

In this paper we shall first define a set of basic social motives; then we shall present some interesting supplementary motives. All the motives will be shown graphically. To clarify the interrelationships among key motives, a correspondence matrix will be considered. By combining some of the basic motives, we find that some useful composite motives emerge. These will be developed and discussed. In the final section, we shall point out how this framework of social motives can be applied to studies of justice, social aggregation, bargaining and decomposed games, subjects which will be treated in subsequent papers.

BASIC SOCIAL MOTIVES AND REPRESENTATION

In order to isolate the structure of social motives, we confine our attention to the simplest social situation, a single decision maker choosing among alternatives which yield consequences to two social units. In such cases the decision maker may concern himself with only the consequences to the first social unit, only the consequences to the second social unit, or the consequences to both. While there are many situations in which a decision maker's choices have consequences or payoffs for social units of which he is not a part, it is particularly interesting to study situations in which his own interests correspond to those of one of the units. Throughout this paper, we shall assume that you, the reader, are the decision maker, and we shall refer to your payoffs and other's payoffs.

When you choose only on the basis of your own payoffs and ignore those of the

other unit, two basic motives can be identified.

Self-interest: choosing so as to increase your own payoffs.

Self-sacrifice: choosing so as to decrease your own payoffs.

Self-interest is a motive that has received considerable attention in economics and game theory. Most tractable models of economic man assume that he is concerned only with his own well-being. Although self-interest can be defined to be "whatever one maximizes," the concept then becomes tautological. When it is defined as above, it becomes more meaningful but of more limited applicability. Self-sacrifice, as defined by this situation, is not done to help others; hence, it must be considered purely masochistic, and abnormal as a motive.

When you choose only on the basis of the payoffs to the other unit and ignore your own payoffs, two additional basic motives can be identified.

Altruism: choosing so as to increase the payoffs to the other unit.

Aggression: choosing so as to decrease the payoffs to the other unit.

The motives are relevant to both human and lower animal social behaviors. Altruism is evidenced by charitable and volunteer activities in man and by helping activities in general. Aggression, as defined here, means hurting the other unit irrespective of the cost to you.

Similarly, when you choose on the basis of both the payoff to yourself and the payoffs to the other unit, we can identify two more basic motives. One form results from your considering the sum of the payoffs while the other results from considering the difference.

Cooperation: choosing so as to increase the sum of your payoff and the other's payoff.

Competition: choosing so as to increase the difference between your payoff and the other's payoff.

The cooperative motive has been the central focus of the utilitarians for over 200 years. Your concern is to make the overall payoff as large as possible, i.e., the greatest good for the greatest number, trusting, perhaps, that an appropriate division can

be arranged later. Competition differs from aggression in that you are trying to get as far ahead of the other unit as possible—you are not just trying to damage the other unit.

It will be convenient to think of the motives in terms of maximizing or minimizing the payoffs with which you, the decision maker are concerned. For convenience, we shall denote your payoffs as x and the other unit's payoffs as y . Cooperation, for example, can then be expressed as $\max(x + y)$.

The definitions we have used in this section do not conflict strongly with those in general usage, but, as we shall see later, the scope encompassed by a motive label such as altruism is very broad. Since it seems useful to deal with pure cases before developing some of the nuances, we have kept our descriptions here extremely simple. The more complex cases will be dealt with later.

GRAPHIC REPRESENTATION OF BASIC MOTIVES

The basic social motives we have considered above can be shown graphically. Along the horizontal axis we shall represent your payoffs (x), and along the vertical axis we shall represent the payoffs (y) to the other social unit. We shall assume that the payoffs increase monotonically along both axes. Any point in the (x, y) plane represents a particular combination of payoffs, a conceivable alternative.

Let us begin by focussing on a particular reference alternative, labelled R , arbitrarily placed in Fig. 1. In order to have a choice, the decision maker must have at least one another alternative. Suppose that you are given a choice of alternative R or some other alternative, A . Which would be preferred under each of the basic motives? If you were self-interested, you would prefer A since it yields you a larger payoff than does R . For the converse reason you would prefer R if your motive were self-sacrificing. If you were altruistic, you would prefer A since it yields the other more than does R . Conversely, you would prefer R if your motive were aggressive. If you were cooperative, you would prefer A since by increasing both your payoff and the other's payoff, it necessarily results in

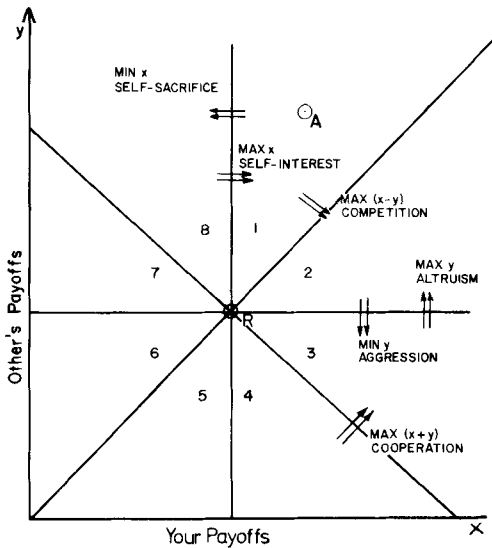


FIG. 1. Preference boundaries of basic motives.

a higher sum than R ; but if you were competitive, you would prefer R because A gives the other unit a larger increment than you receive.

To determine your preference for R versus any other point, not just A , preference boundaries can be established for each of the motives. A preference boundary for a reference point R is defined by partitioning the payoff space into those points preferred to R and those to which R is preferred. For the motive of self-interest, for example, the preference boundary is the vertical line through R since any point to the right of this line gives you a higher payoff than R , while any point to the left of this line gives you a lower payoff than R . This is of course also the preference boundary for self-sacrifice. Under a self-interested motive you would prefer points to the right of the line; under a self-sacrificing motive, you would prefer points to the left of the line. The double arrows in Fig. 1 point in the direction of increased preference for alternatives to R . Fig. 1 also indicates the preference boundaries for the other basic motives. The horizontal line through R is the preference boundary for altruism and aggression, the line with slope $+1$ through R is the preference boundary for competition, and the line with slope -1 through R is the preference boundary for cooperation. These bound-

aries partition the payoff space into eight sectors. All the points within a sector are motivationally equivalent in that if any of the points in a sector is preferred, or dispreferred, to R , under one of the motives considered, then any other point in the sector must be preferred, or dispreferred, to R under the same motive. This congruence will hold for each of the motives used to generate the sectors. On the other hand, for points in different sectors, you can always select at least one of the motives such that one of the points will be preferred to R and one will be dispreferred.

For points that fall on the boundary itself there are two major possibilities: (1) you are indifferent between R and any point along the boundary, and (2) points along the boundary may be preferred or dispreferred to R . For situations in which the first possibility holds, the boundary is called an indifference curve. An indifference curve is a locus of equally desirable points, i.e., you would not care which of the points someone might choose for you from the set of boundary points. The second possibility is the general one and the one we shall find the most useful. When we consider combinations of motives, we shall discover three cases: (i) all the points along the boundary are preferred to R , (ii) all the points along the boundary are dispreferred to R , and (iii) some points are preferred while some are dispreferred. This will be elaborated in the section on combinations of motives.

Clearly, any point in the payoff space could be chosen as a reference point and a preference boundary constructed through it for a particular motive. Hence, for any basic motive there is not one but rather a family of preference boundaries. With a self-interest motive, for example, you would have vertical preference boundaries through all possible reference points, not just through the point R shown. For the other basic motives, the preference boundaries would be the family of straight lines parallel to the single boundary shown in Fig. 1. Families of preference boundaries for some motives to be described in the next section are shown in Fig. 2. A system of preference boundaries based on different

reference points is called a preference structure. When the preference boundaries are indifference curves, the system is called an indifference map. Most of the time we will be concerned with more than one curve but not with the whole map. However, we will want enough of the map to cover those parts of the payoff space that are being studied.

SUPPLEMENTARY BASIC MOTIVES: PROPORTIONALITY

The basic motives identified in the preceding section were restricted to those that can be described in terms of sums or differences of payoffs. This is equivalent to saying that these motives can be represented by a preference structure of parallel straight lines. A first step in expanding this structural form is to consider motives revealed in preferences for products or ratios of the payoffs. The purpose of this consideration is far from mechanistic. The motives so defined are intrinsically interesting and seem to be commonly held. They provide new insight into the meaning of competition and cooperation.

Earlier we defined competition as maximizing the difference between your payoff and the other person's. This definition is implied when, for example, you are concerned about the \$3,000 gap between your colleague's annual salary and your own. Another concept of competition is exhibited, though, when employees compare the percentage wage increase offered in their industry with that received by employees in other industries. Similarly, business firms may focus on market shares rather than on profit differentials; nations may bargain in terms of ratios of missile strength or kill capability rather than in terms of absolute numbers. These situations suggest a new definition.

Proportionate Competition: choosing so as to increase the ratio of your payoff to the other's payoff.

The ratio between payoffs can be maximized by (i) holding the other person's payoff constant while increasing your own, (ii) reducing the other's payoff while holding yours constant, (iii) increasing your own payoff proportionately more as both change, or (iv) reducing the other's payoff

proportionately more as both change. The preference boundaries for proportionate competition are lines with positive slope through the origin. The direction of stronger preference is downward and to the right. These boundaries are shown in Fig. 2 as the fan of straight lines passing through the origin.

In a similar manner we can identify an alternative form of cooperation.

Proportionate Cooperation: choosing so as to increase the product of your payoff and the other's payoff.

The basis for this motive is not as obvious as those for the motives already considered. That is, why you might want to maximize the sum, difference or ratio of payoffs seems apparent; but why you might want to maximize the product does not. As we noted earlier, maximizing the sum of the payoffs implies that the total pie is being increased and that people are or can be made better off with a larger pie. It does not, however, exhibit a concern with the distribution of the pie. But if you know that the total is being made larger, and that you are not going to receive a good share of it, you may not find this form of cooperation desirable. In contrast, a mo-

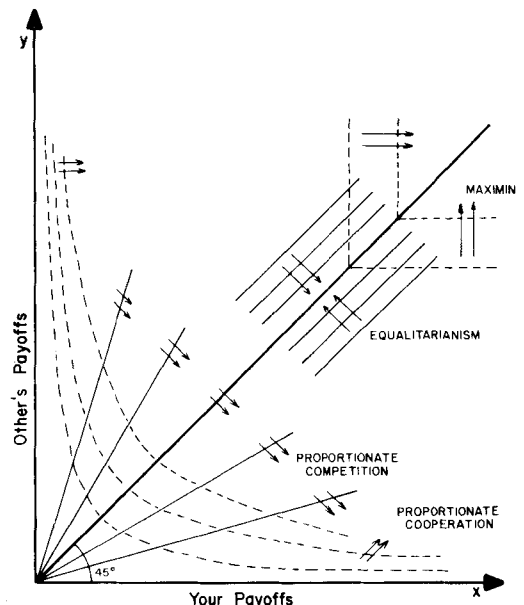


FIG. 2. Preference boundaries for proportionate motives, lower left, and for two conditional motives, upper right.

tive of maximizing the product of the payoffs builds in a concern for both the size of the pie and how it is sliced. With a proportionate cooperation motive, you would not find a bigger pie desirable if your share or the other's share, for that matter were small. With this motive you would, in some circumstances, accept a smaller pie if the shares were more equal than those of a slightly larger alternative. In general, then, we can expect regular cooperation when there is a large pie which will ultimately be redistributed, but when there is no opportunity for later redistribution, we can expect proportionate cooperation. This distinction in forms of cooperation is relevant to the compensation principle in welfare economics. The concept of proportionate cooperation accounts for situations in which the gainers pay the losers but compensation does not actually take place (Kaldor, 1939; Scitovsky, 1941).

The preference boundaries for proportionate competition are rectangular hyperbolas, shown in Fig. 2 by the dashed curves. The direction of increasing preference is up and to the right.

CONDITIONAL MOTIVES AND EQUALITY

Each of the social motives described thus far is constant throughout the payoff space, indicating that the individual's motive is constant regardless of whether his payoffs are greater than, or less than the other's payoffs. Such global uniformity is, however, restrictive, since an individual might exhibit one type of motive when he is ahead and a quite different motive when he is behind. While this type of conditional motive could be characterized in a number of ways, the condition that we will focus on is the position of the payoffs relative to a line of slope +1 through the origin. This equality line is particularly interesting since it is the locus of equal payoffs for you and the other person. If you choose points below and to the right of the equality line, you receive higher payoffs than the other; if you choose points above and to the left of the equality line, you receive lower payoffs than the other. Empirical studies indicate that quite different motives often may operate on different sides of the equality line, i.e., the motive in operation is based on

whether you are ahead or behind (Messick & Thorngate, 1967; McClintock, Messick, Kuhlman & Campos, 1973). The most frequent motive in one experiment was self-interest when behind and cooperation when ahead (MacCrimmon, 1973). Such motives will be called conditional motives.

A conceptually important social motive based on the equality line is the one in which competition holds above the line, while anticompetition holds below the line. This is the motive of equalitarianism.

Equalitarianism: choosing so as to make your payoff and the other's payoff more equal.

This can be stated more precisely as trying to reduce the absolute difference between the payoffs, i.e., minimize $|x - y|$. Although, in general, we might not expect individuals to exhibit a drive toward complete equality which disregards both the consequences to themselves and the consequences to the other person, equalitarianism is a basic motive against which more complex notions of equity or justice can be compared. Fig. 2 shows the preference structure for equalitarianism. Note that in this case there is no ambiguity about choices between alternatives on different sides of the equality line.

Instead of combining competition and anticompetition, we could combine proportionate competition and antiproportionate competition to develop a proportionate equalitarianism. This motive would result in choices which decrease the ratio between the larger payoff to the smaller payoff. For situations in which proportionate competition is more meaningful than regular competition, proportionate equalitarianism, too, may be a more useful concept than regular equalitarianism. That is, when the payoffs are relatively large, proportionate equalitarianism may be more appropriate, but when the payoffs are smaller, regular equalitarianism may operate.

Another conditional motive that has important implications for equality and justice is maximin.

Maximin: choosing so as to make the worse off person better off.

This concept implies that when you are ahead, you become altruistic, but when

you are behind, you become self-interested. Hence it involves a desire to improve the position of the person who is behind but does not necessarily force the person who is ahead to make a sacrifice. While the maximin motive carries some implication of fairness in choices, it still allows the person who is ahead to maintain his lead so long as the other gains something. In a macro context, maximin has been advocated by Rawls (1970) as a motive for justice. The preference boundaries for maximin are shown in Fig. 2.

Even though both equalitarianism and maximin have connotations of justice, note that viewing them as conditional motives reveals their different underlying bases: equalitarianism is composed of competition and anticompetition, while maximin is composed of self-interest and altruism. The components of, and relationships among, all the motives considered thus far will be considered in the next section.

MOTIVATIONAL CORRESPONDENCES

The preference boundaries for the six basic motives initially considered partitioned the payoff space into eight sectors around the reference point. Now that additional motives have been defined, these motives' preference boundaries entail further partitioning of the payoff space. Fig. 3 shows that these subpartitions yield a total of 36 sectors. Each of these sectors contains particular motivational equivalences. For example, compared to *R*, any alternative falling in sector 2a would be preferable if the motive were self-interest, altruism, competition, cooperation, proportionate cooperation, or maximin, but would be dispreferred if the motive were self-sacrifice, aggression, proportionate competition or equalitarianism. These motivational correspondences are summarized in matrix form in Table 1. For each motive listed, a 1 indicates the sectors, i.e., all the alternatives represented by the sector which are preferred to *R*, and a 0 indicates the sectors over which *R* is preferred. Note, then, that reading across the row corresponding to sector 2a gives the division of motives shown above, that is, reading across a row will indicate the possible motives operating for a given choice.

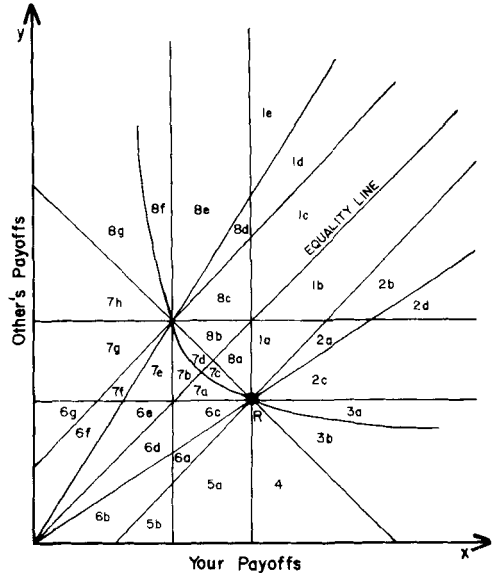


FIG. 3. The partition of the payoff space by the preference boundaries for the motives listed in Table 1.

Reading down a column, e.g., proportionate competition, shows a 1 for sectors 2c, 2d, 3a, 3b, 4, 5a, 5b, 6a, and 6b. These are all the sectors below a line connecting the origin with point *R*, i.e., the preference boundary for proportionate competition. Reading down a column, then, gives all the possible choices which might be made under a given motive. Since some sectors have the same motivational implications for all the motives we are considering here, these sectors have been listed on the same line, e.g., 8a, 8b, and 8c.

The motivational implication matrix of Table 1 can be used to assess the similarity of any two motives in terms of their preference pattern over sectors. This assessment of similarity may be done in various ways. For example, you can compute the correlation between any pair of motives across sectors. Since preferences are binary, this correlation can be obtained by calculating the ϕ coefficient for the 2×2 contingency table formed by constructing the joint prefer-disprefer distribution for a pair of motives (Hays, 1973, p. 743). These correlations are given in Table 2. The similarity of two motives can also be assessed simply by counting the number of sectors for which the two motives imply the same choice. This type of motivational corre-

TABLE 1
MATRIX SUMMARY OF THE INFORMATION IN FIG. 3.

Sectors	Motives									
	Self-Inter- est	Self-Sacri- fice	Altruism	Aggres- sion	Competi- tion	Propor- tionate Competition	Coopera- tion	Propor- tionate Cooperation	Equalitar- ianism	Maximin
1a, 1b, 1c	1	0	1	0	0	0	1	1	1	1
1d, 1e	1	0	1	0	0	0	1	1	0	1
2a, 2b	1	0	1	0	1	0	1	1	0	1
2c, 2d	1	0	1	0	1	1	1	1	0	1
3a	1	0	0	1	1	1	1	1	0	0
3b	1	0	0	1	1	1	1	0	0	0
4	1	0	0	1	1	1	0	0	0	0
5a, 5b	0	1	0	1	1	1	0	0	0	0
6a, 6b	0	1	0	1	0	1	0	0	1	0
6c, 6d, 6e, 6f	0	1	0	1	0	0	0	0	1	0
6g	0	1	0	1	0	0	0	0	0	0
7a, 7b	0	1	1	0	0	0	0	0	1	1
7c, 7d	0	1	1	0	0	0	0	1	1	1
7e, 7f	0	1	1	0	0	0	0	0	1	0
7g, 7h	0	1	1	0	0	0	0	0	0	0
8a, 8b, 8c	0	1	1	0	0	0	1	1	1	1
8d, 8e	0	1	1	0	0	0	1	1	0	1
8f	0	1	1	0	0	0	1	1	0	0
8g	0	1	1	0	0	0	1	0	0	0

Note: For any sector, motive pair, a 1 is entered if any point in the sector is preferred to the reference under that motive. A 0 is entered if, under the motive, the reference is preferred to the sector.

spondence is shown in parentheses in each cell. The correlations in Table 2 should be regarded only as rough approximations to true correlations since they are computed by giving equal weight to each sector despite the obvious fact that the sectors vary widely in size. Obviously, work is needed to refine these estimates and to determine, for example, the dependence of the estimates on the position of the reference point.

Not surprisingly, there is a high correlation between cooperation and proportionate cooperation and between competition and proportionate competition. Looking at the graph representation, Fig. 3, also confirms a strong relationship between proportionate cooperation and maximin. Observe that maximin is positively related to self-interest, but equalitarianism is negatively related to self-interest. It is interesting to note that, contrary to what Rawls (1970, p. 30) seems to argue, maximin implies not only more altruism than does equalitarianism, but also more altruism than does cooperation or utilitarianism.

SIMPLE COMBINATIONS OF MOTIVES

Up to this point we have considered only that type of combination in which one basic motive is applied on one side of the

equality line, and another basic motive is applied on the other side. In this section we shall consider mixtures of motives that operate throughout the payoff space.

Linear combinations

The most obvious kind of mixture is a linear combination. With linear combinations, we are not restricted to the motives whose preference boundaries are given in Fig. 1. We can consider any straight line through R as a possible preference boundary. A line with a positive slope greater than 1, for example, can be thought of as a linear combination of competition and self-interest.

Linear combinations have been studied by Sawyer (1966), Wyer (1969) and Griesinger and Livingston (1973), who form a value function $v(xy) = ax + by$ on the payoff space, where $v(x_0, y_0)$ is the value associated with alternative (x_0, y_0) and a and b are weights. All the basic motives can be represented as special cases of such linear combinations. Self-interest, for example, has $a = 1$, $b = 0$, competition has $a = 1$, $b = -1$, and so forth. By appropriately selecting the parameter values, we can get a line with any desired slope through R . A combination of competition and self-interest would be $a = 1$, $-1 < b <$

TABLE 2
SIMILARITY MEASURES BETWEEN PAIRS OF MOTIVES.

	Self-Inter- est	Self-Sacri- fice	Altruism	Aggres- sion	Competition	Proportionate Competition	Cooperation	Proportionate Cooperation	Equalitarianism	Maxi- min
Self-Interest		-1.00 (0)	.13 (18)	-.13 (18)	.52 (29)	.28 (25)	.59 (28)	.48 (26)	-.36 (12)	.36 (24)
Self-Sacrifice			-.13 (18)	.13 (18)	-.52 (7)	-.28 (11)	-.59 (8)	-.48 (10)	.36 (24)	-.36 (12)
Altruism				-1.00 (0)	-.28 (11)	-.52 (7)	.48 (26)	.60 (28)	0 (18)	.71 (30)
Aggression					.28 (25)	.52 (29)	-.48 (10)	-.60 (8)	0 (18)	-.71 (6)
Competition						.70 (32)	.20 (21)	.07 (19)	-.59 (9)	-.07 (17)
Proportionate Competition							-.07 (17)	-.20 (15)	-.33 (13)	-.33 (13)
Cooperation								.79 (32)	-.33 (12)	+.56 (28)
Proportionate Cooperation									-.11 (16)	.78 (32)
Equalitarian- ism										.11 (20)

Note: The first number in each cell is the ϕ -coefficient or correlation between motives across sectors. The second number is simply the number of sectors out of 36 for which the pair of motives leads to the same choice.

0. The preference structure will be a series of straight lines with slope a/b , with the direction of increasing preference being b/a .

While linear combinations generalize the basic motives, they are constraining because they require the same hybrid motive to hold throughout the space, and they do not add any intuitively interesting motives. That is, a motive combining competition and self-interest is just that, rather than being interesting in its own right. Experimental studies by MacCrimmon (1973) and McClintock, Messick, Kuhlman and Campos (1973) suggest that linear combinations may be too restrictive to account for even simple preference data.

Conjunctive combinations

A different kind of combination of motives is formed by conjunction, i.e., joining the two motives by the connective and. A conjunctive combination of self-interest and altruism would require that for any alternative to be preferred to a reference alternative R , it must yield higher payoffs both to you and to the other person. This combination corresponds to any point in sectors 1 and 2 of Fig. 1 being preferred to R , while all points in sectors 3 through 8 are dispreferred. The well known Pareto principle of welfare economics, which re-

quires that everyone be made better off, is this type of conjunctive combination.

The sectors which will be preferred under a particular conjunctive combination can be easily obtained from the matrix of Table 1. By multiplying the columns of motives being conjoined, we obtain products which indicate the preference sectors for the conjunctive motive. For example, by multiplying the self-interest and altruism columns, we obtain a column with 1s for sectors 1a, 1b, 1c, 1d, 1e, 2a, 2b, 2c, and 2d. Sector preferences for any other conjunctive combinations can be obtained in a similar way.

Disjunctive combinations

Forming conjunctive combinations obviously suggests the possibility of forming disjunctive combinations, i.e., joining the two motives by the connective or. A disjunctive combination of self-interest or altruism would require that, for any alternative to be preferred to a reference alternative, R , it must make either you or the other person better off. This combination corresponds to any point in sectors 1-4, 7, and 8 of Fig. 1 being preferred to R while all points in sectors 5 and 6 are dispreferred.

The sector preferences for disjunctive combinations also can be ascertained from

Table 1 by adding (where $1 + 1 = 1$) the columns for the motives being disjoined. For example, the disjunction of self-interest and altruism will have 0s in rows 5a, 5b, 6a, 6b, 6c, 6d, 6e, 6f, 6g, and 1s elsewhere.

Lexicographic combinations

Lexicographic combinations, the final type of simple combinations we shall consider, involve a primary motive and a secondary motive. Let us consider a primary motive of self-interest and a secondary motive of altruism. This could be depicted by placing arrows along the preference boundaries of the primary motive such that the arrows pointed in the direction of preference for the secondary motive. The alternatives are first evaluated in terms of the primary motive, which in our example is self-interest. If this motive leads to a clear choice, the secondary motive is not considered. However, if the alternatives are on the preference boundary of the primary motive, then the secondary motive, altruism in this case, will operate in making the choice. For the self-interest-altruism lexicographic combination, points to the right of the line through a reference point, or points directly above the reference point, are preferred. The choices would be the same for any secondary motive that correlates positively with altruism in that part of the payoff space.

As indicated above, a lexicographic preference structure can help to clarify the ambiguities of choosing between alternatives located along a preference boundary. While one approach would be to assume that the decision maker were indifferent to all the points along a boundary, this assumption seems very unreasonable for most cases. For example, even if someone had equalitarian motives, it is highly unlikely that he would not care whether his objective of equality were realized by making everyone equally poor or by making everyone equally rich. This notion of applying both primary and secondary motives might be extended to all equality line conditional motives. The motive on the opposite side of the equality line could act as the secondary motive. Even for very complex combinations of motives, lexicogra-

phy can usefully clarify the motives along the preference boundaries. For a detailed discussion of lexicographic preferences, including the well-known result that such orderings cannot generally be represented by real valued functions, see Fishburn (1974).

RECONSIDERATION OF BASIC MOTIVES

Now that we have considered types of combinations, let us return to the basic motives presented at the outset of this paper to see if any of them can be usefully elaborated. Since we have focused only on simple two-person cases, we shall consider briefly the extent to which these motives can be easily generalized to n -person cases.

Self-interest

Although self-interest is a very important motive, it is straightforward. Even in n -person cases, it is well defined since, by definition, it ignores external considerations.

Self-sacrifice

Alone, the motive of self-sacrifice is of little interest, but it is useful in combination with other motives, as we shall see below.

Altruism

What altruism implies has been controversial. Consider the sectors in Fig. 1. Would a choice of any point in sectors 1, 2, 7 or 8 imply altruistic behavior? Or should only a choice of points in sectors 7 or 8 imply altruistic behavior? Or would a compromise notion of points in sectors 1, 7 or 8 be appropriate? The sectors 1, 2, 7 and 8 correspond to our original definition of altruism as any choice which improves the payoffs to the other person. The main objection to this definition occurs, however, when we consider a point in the rightmost part of sector 2 and ask how someone who takes a large increment in payoff for himself can be called altruistic, even though the other person benefits a little. The intermediate case, i.e., sectors 1, 7 or 8, involves the conjunction of altruism and anticompetition and suggests that although you can benefit yourself and still be consid-

ered altruistic, the other person has to benefit more. The more extreme definition, which corresponds to choices in sectors 7 or 8 only, requires a definite self-sacrifice while benefitting the other. This is the common view of altruism in evolutionary theory (Trivers, 1971; Campbell, 1972). A more sophisticated concept would take account of the equality line, and your position relative to the other's. Altruism would entail self-sacrifice if you were at an advantage, but would require only an improvement for the other if you were at a disadvantage.

In extending altruism to n -person situations, we can still define it as maximizing other, but for what function of other? Suppose that some people gain and some lose. Is the choice altruistic or not? In the extreme case in which everyone else gains and you lose, n -person altruism is well defined. In the intermediate cases, a more careful specification is needed. Perhaps in n -person cases, as in two-person cases, altruism should be thought of as being directed toward a particular other, or a specific set of others, and one should not talk of general altruistic behavior.

Aggression

As the opposite of altruism, aggression involves many of the same considerations. For instance, can you really be considered aggressive if you opt for a big gain for yourself while hurting the other only a little? Does aggression entail self-sacrifice? Because the other factors pertinent to the altruism discussion apply here, they will not be repeated.

Cooperation

We have considered two forms of cooperation: the basic cooperation and proportionate cooperation. Examining Fig. 3 shows that these motives result in different preferences for sectors 7c and 7d. Under proportionate cooperation, alternatives in these sectors are accepted. Under basic cooperation, alternatives in these sectors are rejected. The alternatives in these sectors decrease the total pie, but they obtain a more equal distribution without sacrificing much of the total. Proportionate cooperation would also reject al-

ternatives in sectors 3b and 8g because, although they increase the total pie, they are distributed very unequally.

As mentioned earlier, both concepts of cooperation are reasonable. It might, in fact, be desirable to combine them conjunctively and disjunctively. Conjunctive cooperation would require that an acceptable point maximize both $(x + y)$ and $(x \cdot y)$. The acceptable points then correspond to regions 1, 2, 3a and 8a-f. These acceptable points require the total payoff to increase, but will not allow either party's share to get too low. Disjunctive cooperation, requiring an improvement in either $(x + y)$ or $(x \cdot y)$, corresponds to points in sectors 1, 2, 3, 7c-d, 8a-g and so would not be concerned with disproportionate shares as long as the total increased. It would, however, allow the total to decrease slightly for the sake of equality. If the distribution of shares is desired in a concept of cooperation, then one of these supplementary forms seems useful. Some people may even want to require that some self-sacrifice occur in truly cooperative behavior, but since the effect of this criterion was considered in our discussion of altruism, where it seems more appropriate, it will be omitted here.

Cooperation has a very important n -person form, utilitarianism. If all the payoffs are commensurate, then the notion of maximizing the total payoff, the greatest good for the greatest number is well defined.

Competition

In discussing proportionate competition, we highlighted the differences between basic competition and the proportionate form, but we might summarize here the sector differences. Proportionate competition would accept, but basic competition would reject, sectors 6a and 6b, while the reverse would be true of sectors 2a and 2b. In the same way that we defined a conjunctive cooperation, we could define a conjunctive competition. It would prefer over R only those alternatives that increased both the difference and the ratio of the payoffs; these preferred points are located in sectors 2c, 2d, 3a, 3b, 4, 5a, and 5b. Thus, increasing the difference while the

ratio narrowed, as do alternatives in sectors 2a and 2b, would not be acceptable, nor would increasing the ratio while the difference narrowed, as do alternatives in sectors 6a and 6b. Disjunctive competition would require that either the ratio or the difference improve and so would add sectors 2a, 2b, 6a, and 6b to the list of conjunctive competition sectors above. Obviously, disjunctive competition is weaker than conjunctive competition.

Equality, justice, fairness

While we have considered some notions of equality, justice and fairness in this paper, the subject is an important one and too long to develop here. It will be the subject of a separate paper.

COMPLEX COMBINATIONS OF SOCIAL MOTIVES

The combinations of motives considered thus far have been easy to specify. There is no reason to expect, however, that a person's actual preference boundary will have a regular shape. Complex curvilinear boundaries such as the family shown in Fig. 4 could arise easily. In fact, empirical studies by MacCrimmon and Siu (1974) have discovered similar preference structures.

Let us examine the implications of your having a preference boundary such as abc . Around point a you are competitive; as the relative advantage of the other person decreases, around point b , you become more self-interested; as you gain the advantage, around point c , you become more cooperative. If we examined a large segment of the curve, we might be able to observe aggression above point a and altruism below point c . This, then, is a complex combination of motives that has no easily expressible functional form.

Suppose point a represents the current state, in which the other person is ahead. Note that you are willing to sacrifice some of your payoff to reduce the payoff to the other person. You are willing, though, to sacrifice only $x_a - x_b$ units at most, and only if you can reduce the other's payoff $y_a - y_b$ units. For any reduction to the other person different from $y_a - y_b$ you are willing to trade off only some amount less than $x_a - x_b$. Although you are willing to

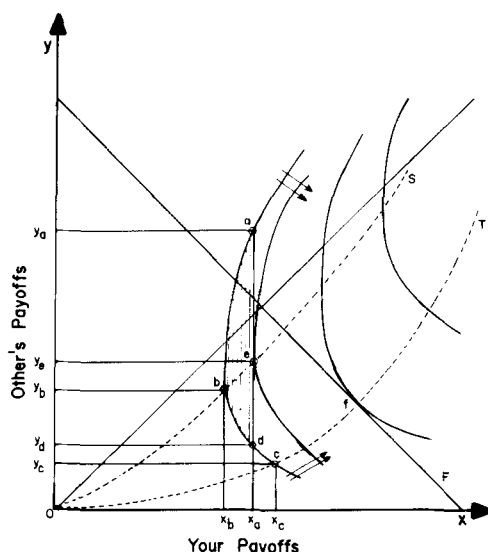


FIG. 4. An example of a complex, curvilinear preference structure.

see the other's payoffs reduced even more, and would, in fact, give up something if they were reduced by any amount up to $y_a - y_b$, the amount $y_a - y_b$ is the ideal amount of reduction or punishment from reference point a . The amount $x_a - x_b$ may be thought of as the intensity of your desire to reduce the other's payoff or punish the other since it is the maximum you would pay. On the other hand, you do not want to see the other's payoffs reduced below y_b unless you are compensated further. This example, then, suggests an appealing method of representing aversion or attraction with respect to another's payoffs. We commonly assume that attractive outcomes are those one would pay for, i.e., accept a reduction from the status quo for, and that aversive ones are those one would have to be compensated for to accept. With this assumption in mind, we can say that the payoffs to the other from y_b to y_a are attractive to you since you would accept any point in the shaded region in preference to the reference a and since these points all involve a reduction in your payoff. Payoffs to the other outside of the interval (y_b, y_a) are, on the other hand, aversive to you since you would only accept them if your own payoff was increased, i.e., if you were paid, too.

From the preference boundaries of Fig.

4, we can also infer your preferred distributions of payoffs. Suppose that your payoff is fixed at x_a , and you have the freedom to establish what the other person will receive. You will select an amount y_c since it allows you to achieve your highest (rightmost) preference curve, the curve tangent to a vertical line through x_a . By taking other possible values at which your payoff could be fixed, we trace out a locus OS of preferred payoffs to the other person facing a payoff determined by you. That this locus OS falls on the advantage side of the equality line suggests that you are motivated by a macro kind of competition.

If instead of facing a fixed payoff to yourself, you were confronted with a fixed total F , shown in Fig. 4 by the line with slope -1 , to be distributed between you and another, we would arrive at a different locus. The most preferred distribution with a fixed amount F is the point f , which is on the highest feasible preference curve. By tracing out all points tangent to a line of slope -1 , we obtain the locus OT of preferred distributions, given a fixed total. If you did not become cooperative, then, of course, the highest preference boundary would be at the abscissa. The locus OT would then lie along the abscissa, indicating that you would prefer to take all the payoff available for distribution.

Finally, if the payoff to the other person were fixed and you then were asked to specify a preferred amount for yourself, the preference map seems to indicate that you would take as much as possible. For the amount to be limited, the preference boundaries would have to be tangent to a horizontal line. Such tangencies are not shown in this figure. For another discussion of these issues, see Scott (1972).

EXTENSIONS

In this paper we have tried to outline some concepts underlying a theory of social motives. Currently, there is considerable controversy about what particular social motives entail. For example, see the exchange between Krebs (1971) and Hebb (1971) about the nature of altruism. We have tried to examine the components of motives at an abstract level in order to make the similarities and differences more

apparent. The need for clarification and operationalization is particularly acute for complex motives. For example, of all the studies dealing with inequity, not one has attempted to ascertain the underlying preference structure. Not a single study discussed in the review by Walster, Berscheid and Walster (1973) is concerned with the measurement of equity motives or their discrimination from the other motives with which they may be confounded. Similarly, Rawls (1970) discusses at considerable length concepts of justice based on utilitarianism, maximin, equalitarianism, altruism, etc., without clarifying the interrelationships among these motives.

The framework that we have described not only permits the conceptual analysis of the interrelationships of various motives, it also has important implications for empirical research. Foremost among these implications are those pertaining to the measurement of social motives. How can one determine what motive underlies an individual's choices?

To begin with, one must in many cases specify the set of feasible alternative motives one wishes to discriminate. For example, if one assumes that the motive will be one and only one of the ten listed in Table 1, the task becomes rather simple. One needs only to select a subset of the sectors and then an arbitrary point within each sector such that no two columns in the matrix formed by the motives, as columns, and the subset of sectors are identical. This is naturally the case with the complete matrix, but it is also the case with the matrix whose rows are sectors 1, 2a, 2c, 3a, 4, and 8g, for example. Thus six binary choices are capable of discriminating perfectly among the ten motives in Table 1.

It would probably be unusual in practice to want to discriminate among so many different motives. Tajfel (1970) considers only four, for example, whereas Messick and McClintock (1968) studied just three. In the former case, the motives were, with a slight redefinition of the axes, self-interest, cooperation, equalitarianism, and competition. These four motives can be discriminated with just two choices. Pairing the reference against a point in sectors

1d, 1e and 4, for example, gives a distinct choice profile for each motive.

In some applications, one may suspect that the motives will be more complex than those listed in Table 1. In such cases, one can attempt to directly estimate the preference boundary for a particular reference alternative. One such effort has been described by MacCrimmon and Siu (1974). In general, such a procedure exploits a very useful property of preference boundaries. Such boundaries partition the outcome space into two mutually exclusive regions, one of which contains all the points preferred to the reference while the other contains those points to which the reference is preferred. Thus by sampling points and observing the choices between these points and the reference, one can estimate the border between the preferred and dispreferred regions, which is, by definition, the preference boundary. This procedure, while quite general, can be quite time consuming without the aid of a computer algorithm for selecting the points.

A wide variety of other measurement procedures can be employed, depending on the nature of the research questions being attacked and the level of specificity desired. Rankings of points selected from the space, for example, will allow an experimenter to determine which of a set of plausible motives are consistent with an observed ordering. Questions like those discussed in conjunction with Fig. 4 can be employed to learn the characteristics of a preference structure. Most psychological studies of equity, for example, use the distribution of a fixed quantity of some sort as the dependent variable. As was noted in our discussion of Fig. 4, the answer to any such single question provides at best only partial information about the underlying motive.

While the framework we have described not only allows one to pose questions regarding social motives clearly and to devise measurement techniques to answer these questions, it suggests in addition a number of important questions.

First and foremost, the framework suggests that there may be a wide variety of more or less complex ways of evaluating or judging social distributions of outcomes. A

question of psychological significance is the degree to which such motives are situationally, as opposed to dispositionally, determined. While this is not the place for a review of the pertinent literature, we do not want to leave the impression that when discussing a motive like altruism, for example, we have in mind a stable personality characteristic of a person. On the contrary, we tend to believe that the motives we have discussed are in large measure situational and influenced by a variety of aspects of the interpersonal relationship. It is important therefore to understand these influences.

Regardless of one's position on the situational-disposition issue, the fact remains that social motives are intraindividual in that they characterize one person's preferences. Given an interpersonal situation, we can assess social motives by studying the choices or preferences of a single person. Our interest and that of most social scientists, extends beyond the study of single individuals to the study of the forms of interdependence that characterize the social system in which we live, however, and it is in this enterprise that the study of social motives is most urgently needed. We know, both from logic and empirical studies (Deutsch, 1958; Griesinger & Livingston, 1973), that some pernicious forms of social interdependence, i.e., the prisoners' dilemma, cease to be problematic if the persons are cooperatively motivated rather than self-interested. There is evidence, on the other hand, to suggest that in certain bargaining contexts, cooperatively motivated persons achieve poorer outcomes than self-interested persons (Kelly & Schenitzki, 1972). In the context of prisoners' dilemma research again, both McNeel (1973) and Kuhlman and Marshello (1975) have shown that differently motivated persons respond radically differently to various strategies of the other. And in political economics, it has long been known that the problem of cyclical group preferences, another worrisome form of social interdependence, evaporates if certain restrictions, i.e., single peakedness, are placed on the individual preferences (Sen, 1970).

What these bits of knowledge indicate is

that the consequences of various forms of interdependence can be quite different depending on the nature of the preferences of the people involved. There are, unfortunately, rather large gaps in our knowledge of the principles that relate individual motives to collective outcomes. To the extent that this framework helps to clarify and consolidate what we know about social motives or provides useful guidelines for the assessment of such motives, to that extent it will help plug some of these gaps.

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